APM with the Elastic Stack

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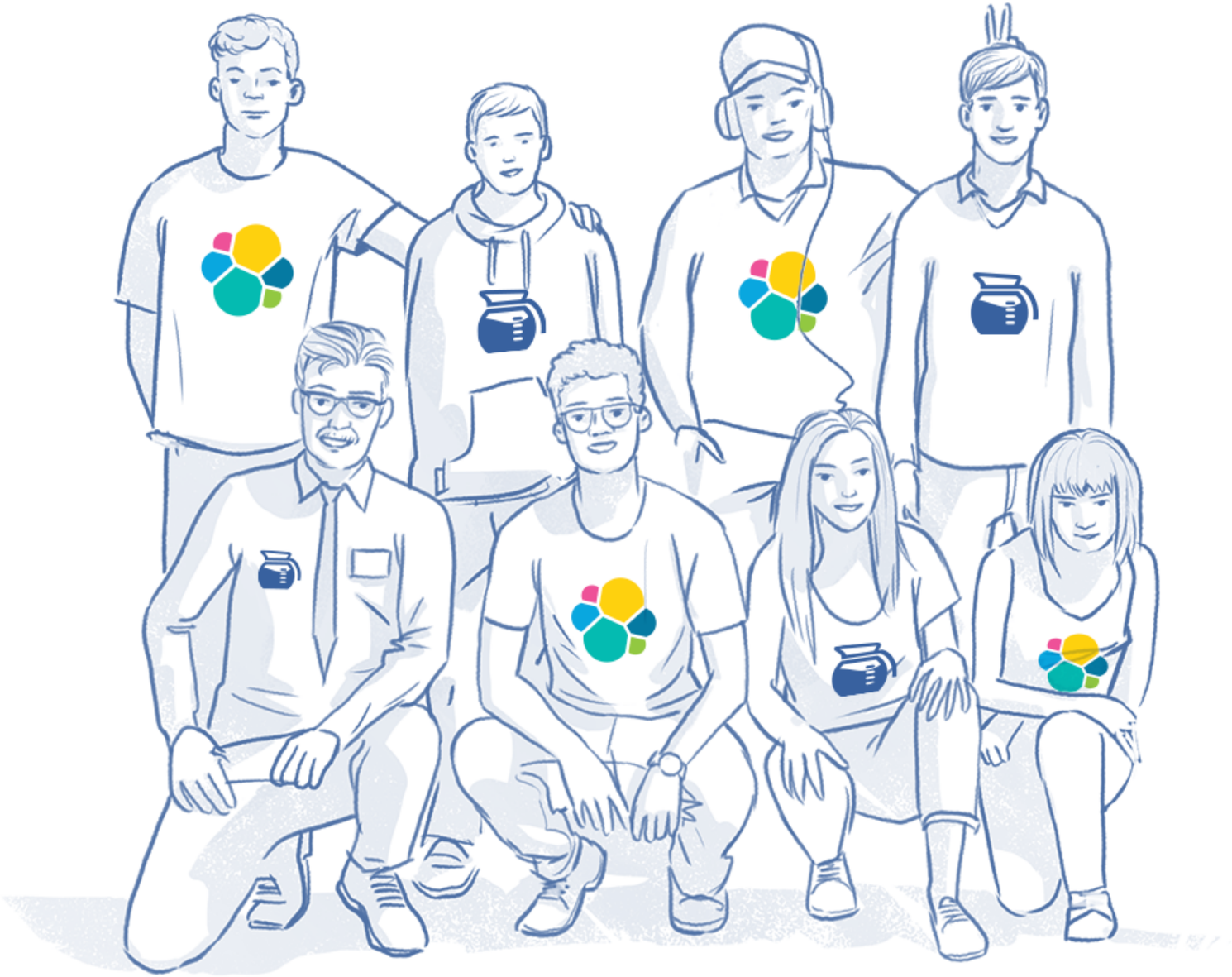
Ron Cohen, Tech Lead, APM

Elastic

February 28, 2018

@makwarth @roncohen



What is APM?

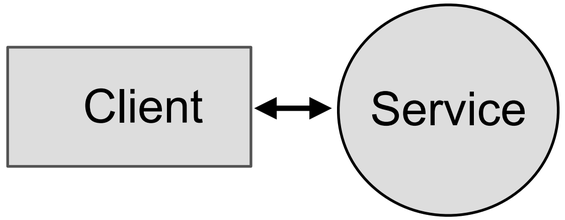
{ }

Application Performance Monitoring (and management) of ***performance and availability of software applications***

{ }

APM helps IT and developers ***monitor the application layer*** *of the stack and the* ***end-user experience***

Problem

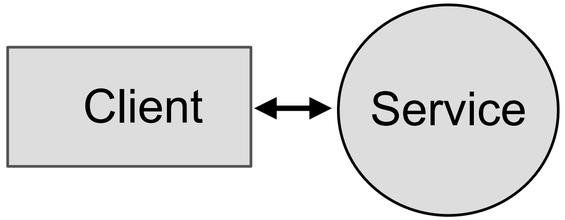



```
08:32:10 Request “/api/products/top”
```

```
08:32:17 Response “/api/products/top 200 OK”
```

7 seconds - zzZZzzZz

Problem

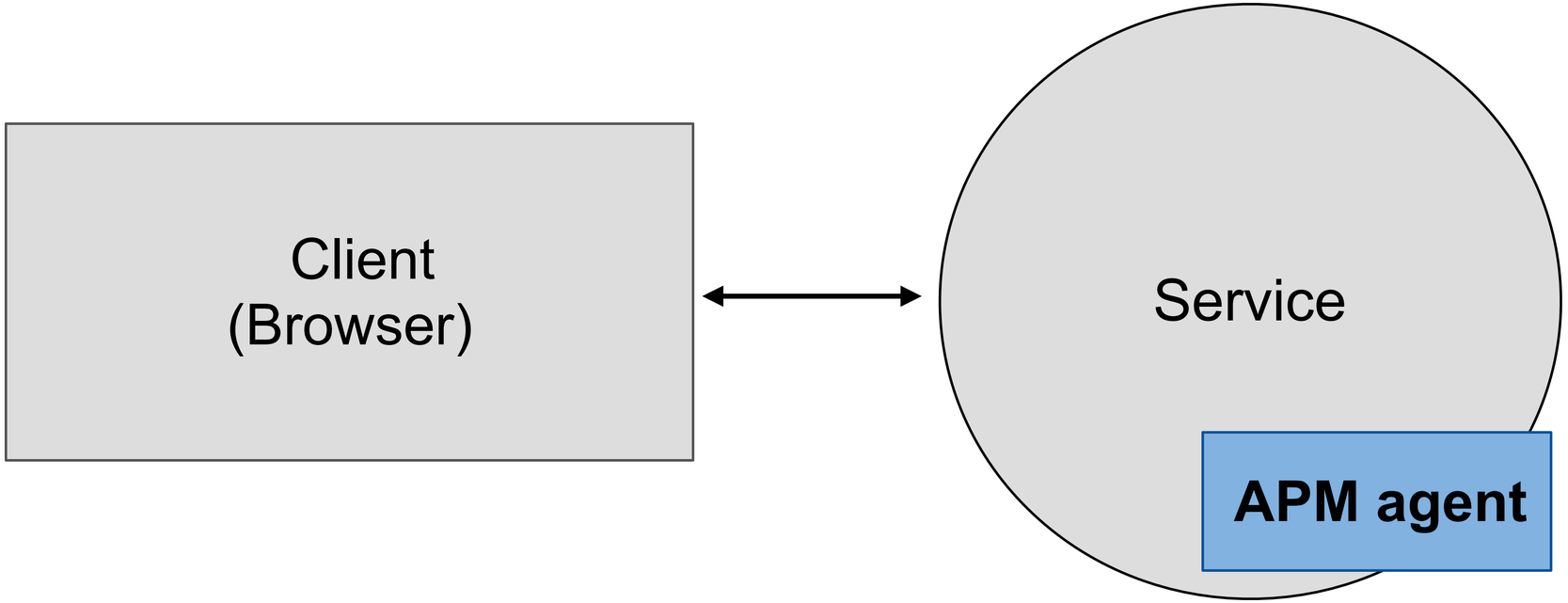



```
08:32:10 Request “/api/checkout”
```

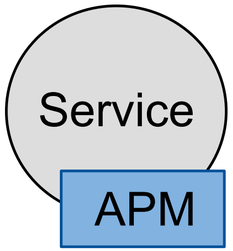
```
08:32:17 Response “/api/checkout 500 ERROR”
```

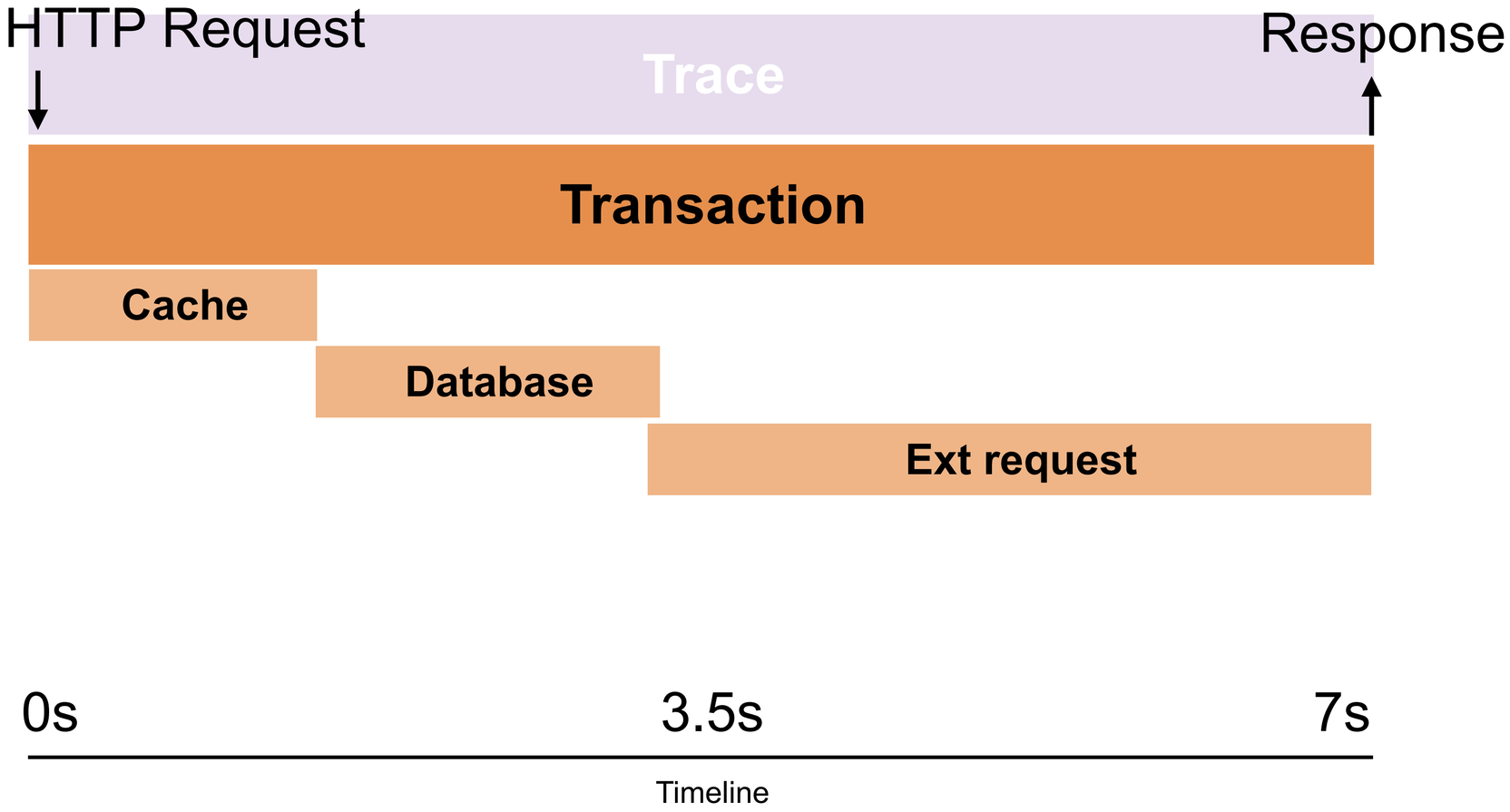
APM agents

Agents are libraries in your applications



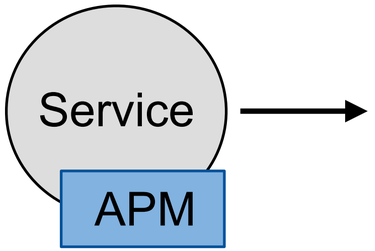
GET /api/products/top

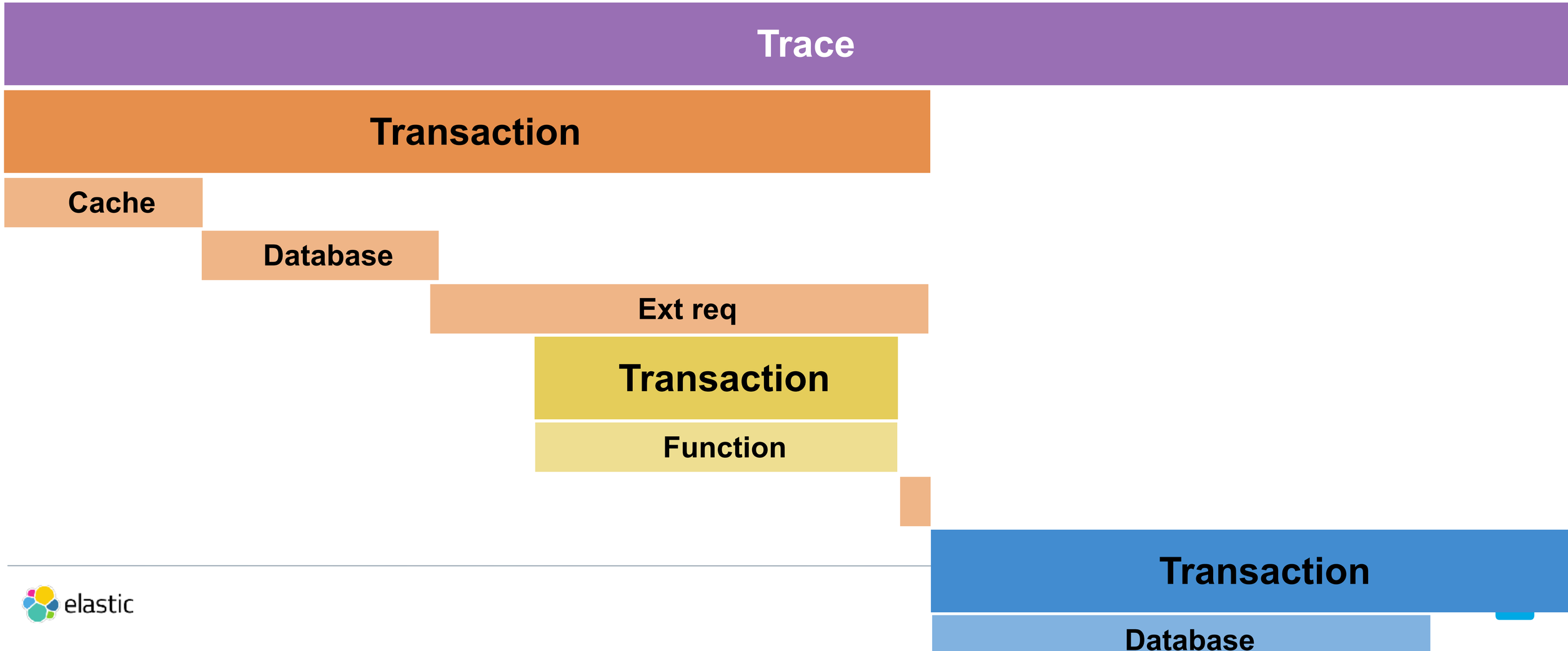




GET /api/products/top

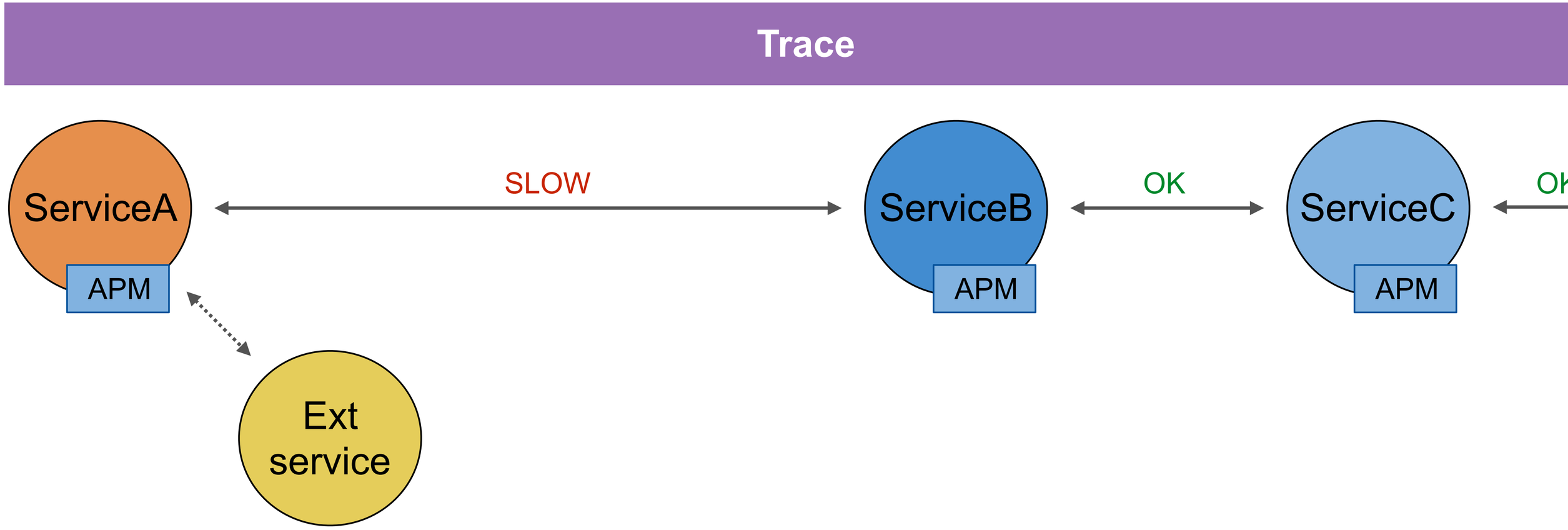
A trace can span multiple services

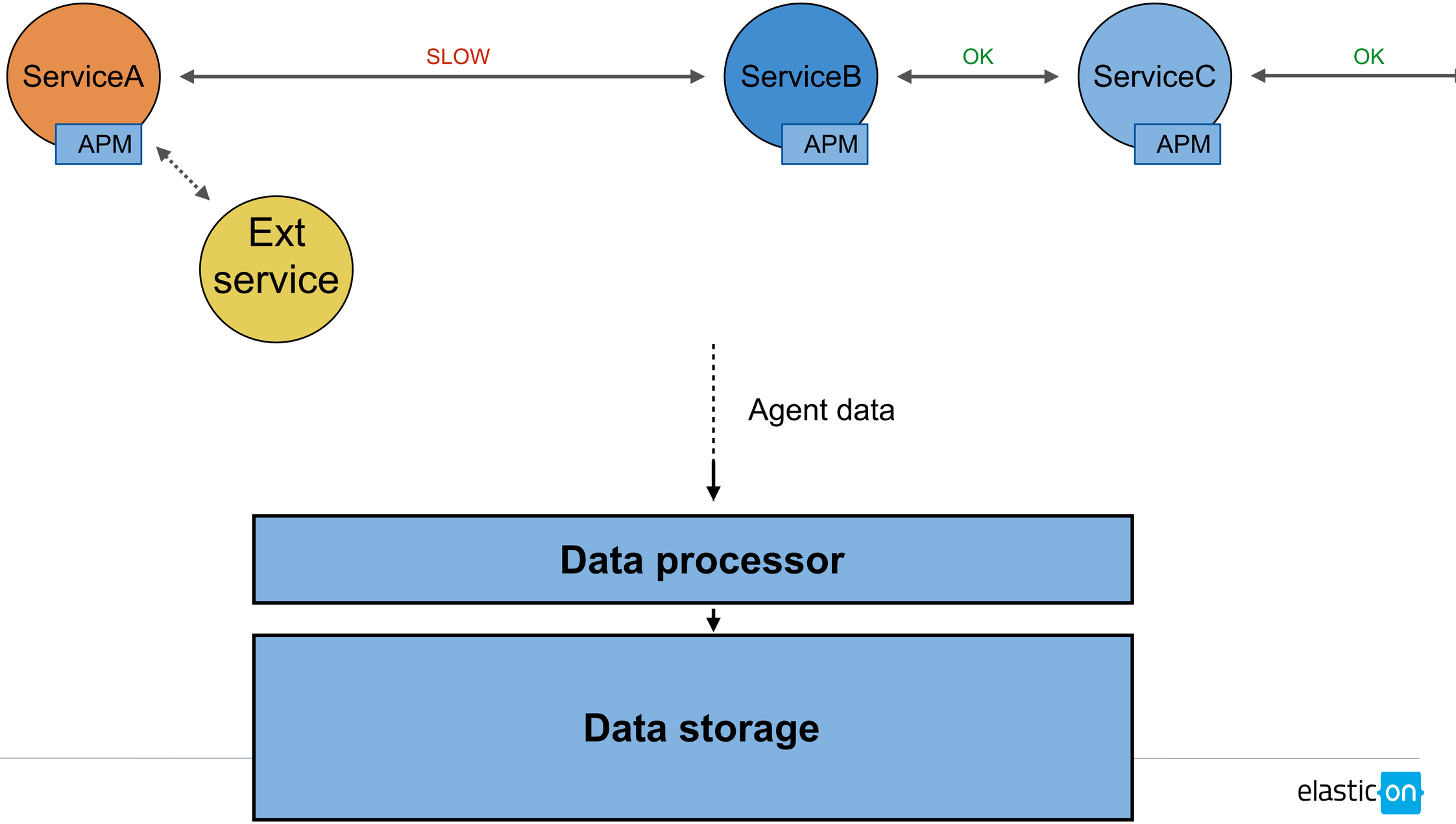




GET /api/products/top

Distributed tracing




ServiceA
APM
SLOW
ServiceB
APM
OK
ServiceC
APM
OK
Ext service
Agent data
Data processor
Data storage

- Understand performance impact of code changes
- Be proactive
- Resolve issues faster

Introducing

Elastic APM

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Elastic APM is GA

Already housing logs and metrics in Elasticsearch? Now add APM.

Attend Webinar Learn More

Deploy Hosted Elasticsearch

Spin up a cluster with Elastic Cloud.

Learn More

Getting Started with Kibana

Create your first dashboard.

Watch Now

Looking for Support?

We listen. We solve. We guide. We love crushing response times.

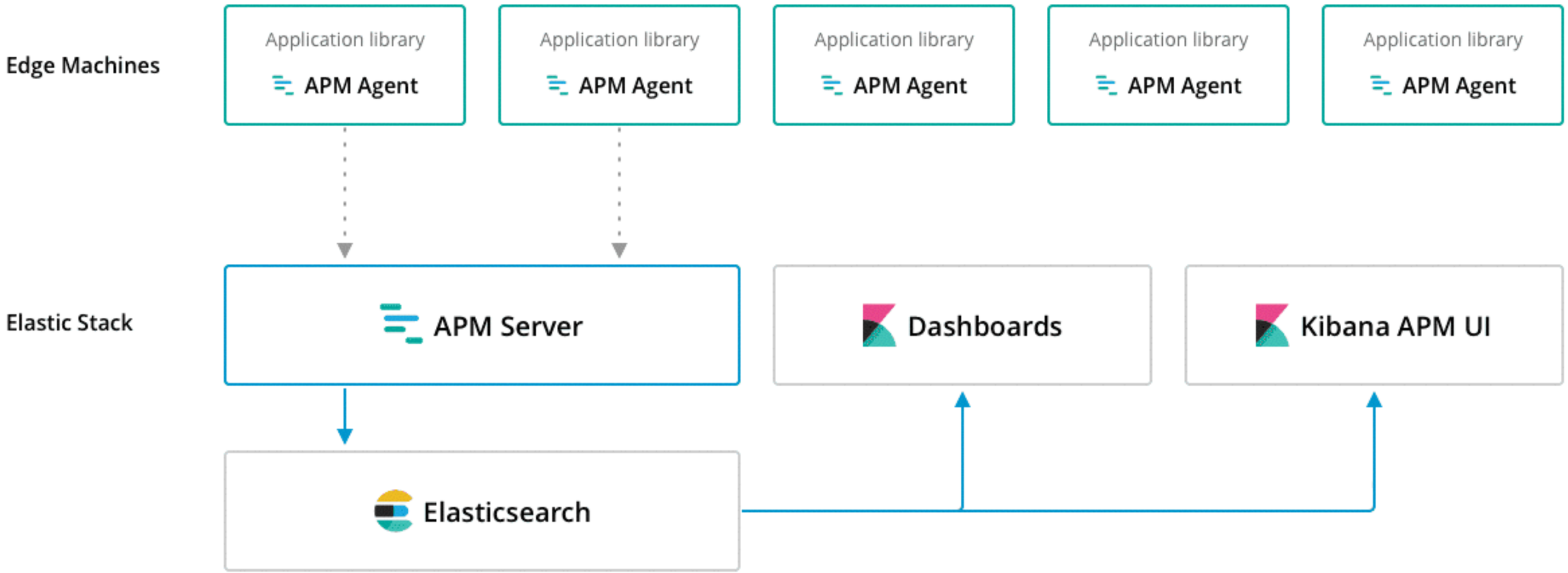
Contact Us

Full-stack monitoring with Elastic

APM adds end-user experience and application-level monitoring to the stack

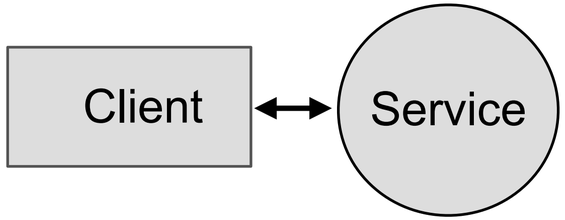
APM	Real User Monitoring (RUM)
APM	Application-level monitoring
Beats	Server-level monitoring
Beats, Logstash	Logging

Setup


Edge Machines
Application library
APM Agent
Application library
APM Agent
Application library
APM Agent
Application library
APM Agent
Application library
APM Agent
Elastic Stack
APM Server
Dashboards
Kibana APM UI
Elasticsearch

Demo!

Real User Monitoring (RUM)



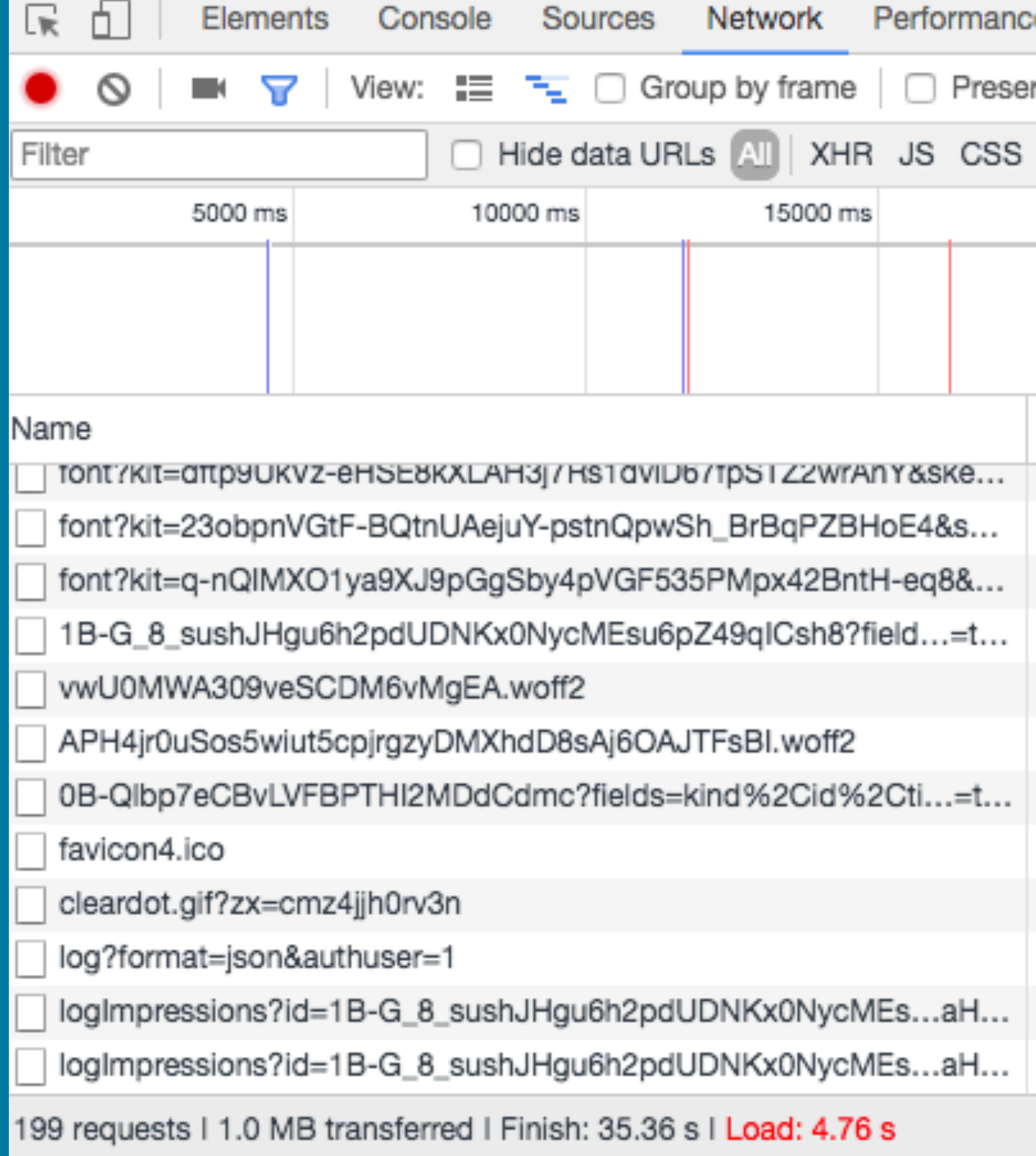

```
08:32:10 Browser: Request “GET /api/products/top”
```

```
08:32:11 Server response “/api/products/top 200 OK”
```

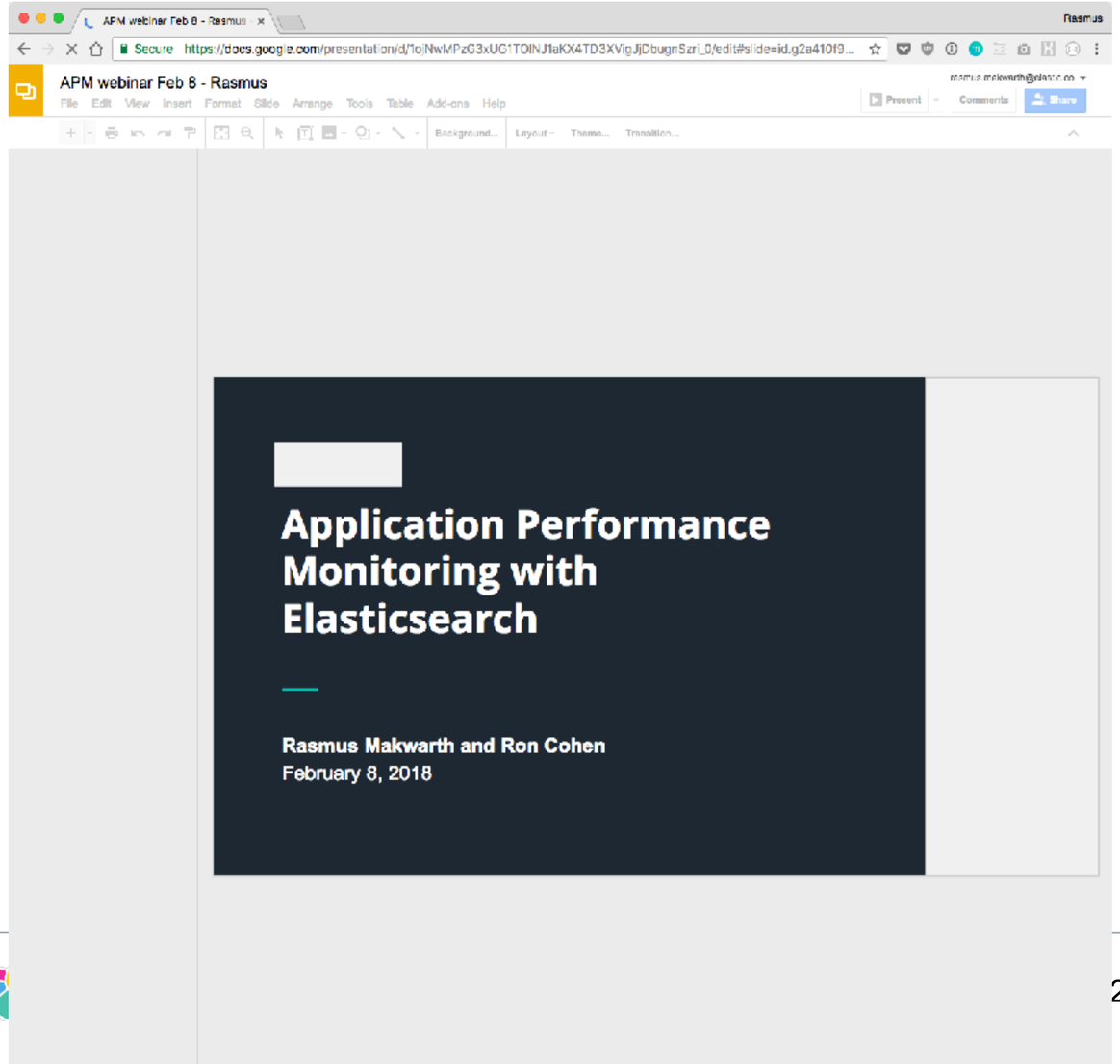
```
08:32:17 Browser: Time To Interaction
```

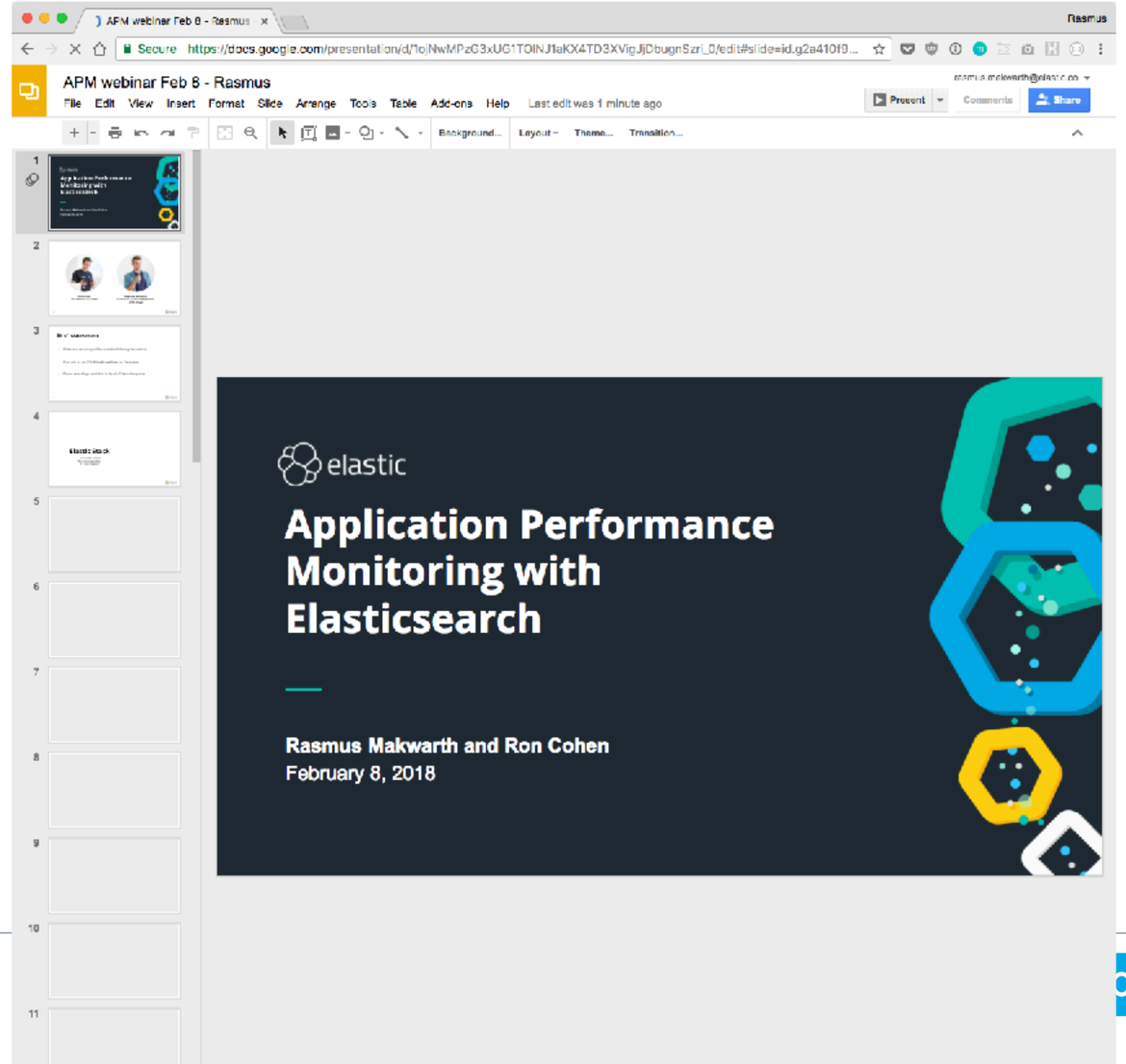
5 seconds

Actual end-user experience can only be measured in the client



Time To Interaction: 17 secs





Elements | Console | Sources | Network | Performance | Memory | Application | Security | Audits

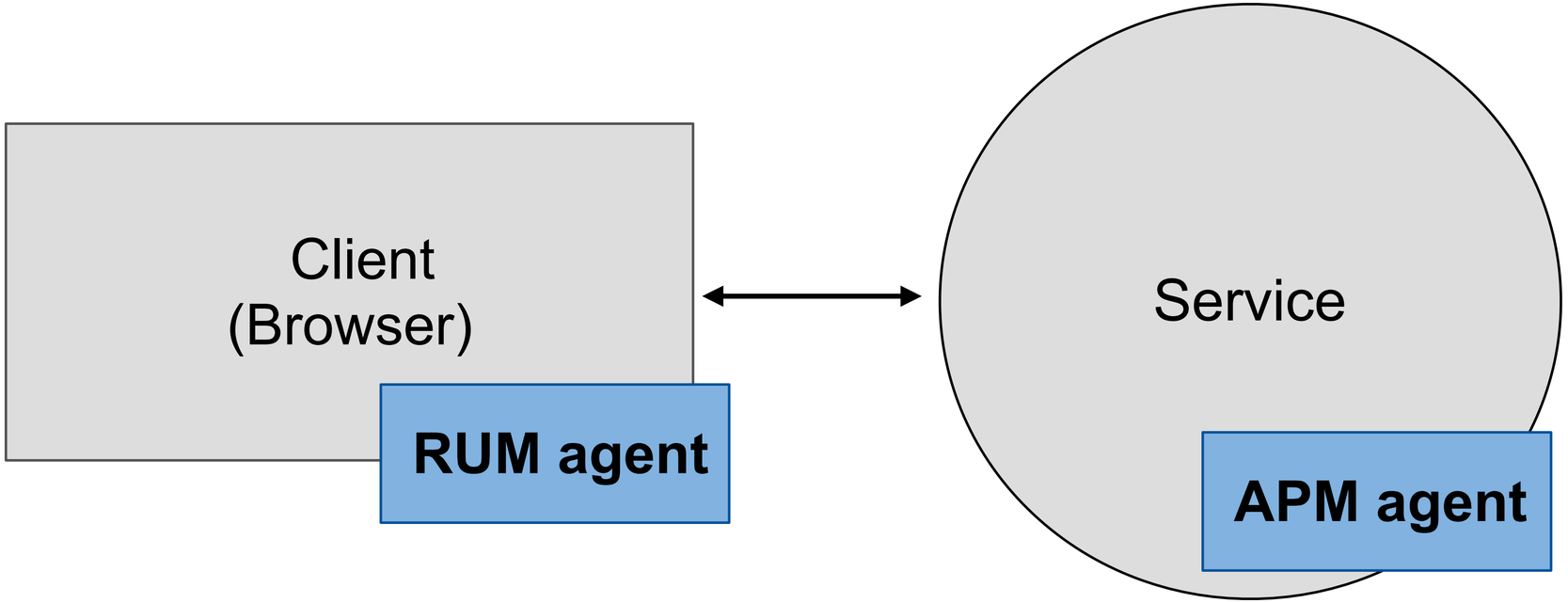
View: | Group by frame | Preserve log | Disable cache | Offline | Online

Filter | Hide data URLs | All | XHR | JS | CSS | Img | Media | Font | Doc | WS | Manifest | Other

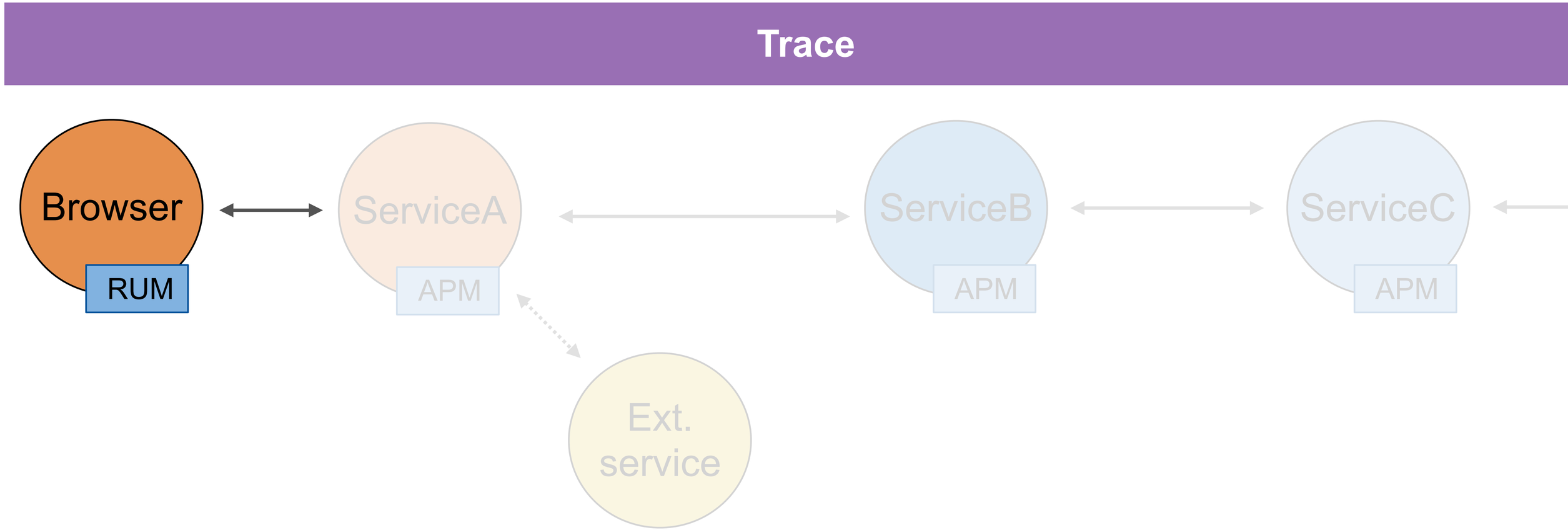
5000 ms | 10000 ms | 15000 ms | 20000 ms | 25000 ms | 30000 ms | 35000 ms | 40000 ms | 45000 ms | 50000 ms | 55000 ms

Name	Status	Type	Initiator	Size	Time	Waterfall
font?kit=dftp9UkVz-eHSE8kXLAH3j7Hs1dvlDb7fpST2ZwrAnY&ske...	200	font	2835220864-editor_core.js:791	(from disk cache)	141 ms	
font?kit=23obpnVGtF-BQtnUAejuY-pstnQpwSh_BrBqPZBHoE4&s...	200	font	2835220864-editor_core.js:791	(from disk cache)	144 ms	
font?kit=q-nQIMXO1ya9XJ9pGgSby4pVGF535PMpx42BntH-eq8&...	200	font	2835220864-editor_core.js:791	(from disk cache)	147 ms	
1B-G_8_sushJHgu6h2pdUDNKx0NycMEsu6pZ49qICsh8?field...=t...	200	xhr	cb=gapi.loaded_0:149	713 B	449 ms	
vwU0MWA309veSCDM6vMgEA.woff2	200	font	2835220864-editor_core.js:791	(from disk cache)	13 ms	
APH4jr0uSos5wiut5cpjrgzyDMXhdD8sAj6OAJTFsBI.woff2	200	font	2835220864-editor_core.js:791	(from disk cache)	12 ms	
0B-Qlbp7eCBvLVFBPTHI2MDdCdmc?fields=kind%2Cid%2Cti...=t...	200	xhr	cb=gapi.loaded_0:149	515 B	330 ms	
favicon4.ico	200	x-icon	Other	193 B	151 ms	
cleardot.gif?zx=cmz4jjh0rv3n	200	gif	2835220864-editor_core.js:425	111 B	24 ms	
log?format=json&authuser=1	200	xhr	edit:119	328 B	219 ms	
logImpressions?id=1B-G_8_sushJHgu6h2pdUDNKx0NycMEs...aH...	204	xhr	2835220864-editor_core.js:345	319 B	481 ms	
logImpressions?id=1B-G_8_sushJHgu6h2pdUDNKx0NycMEs...aH...	204	xhr	2835220864-editor_core.js:345	317 B	301 ms	

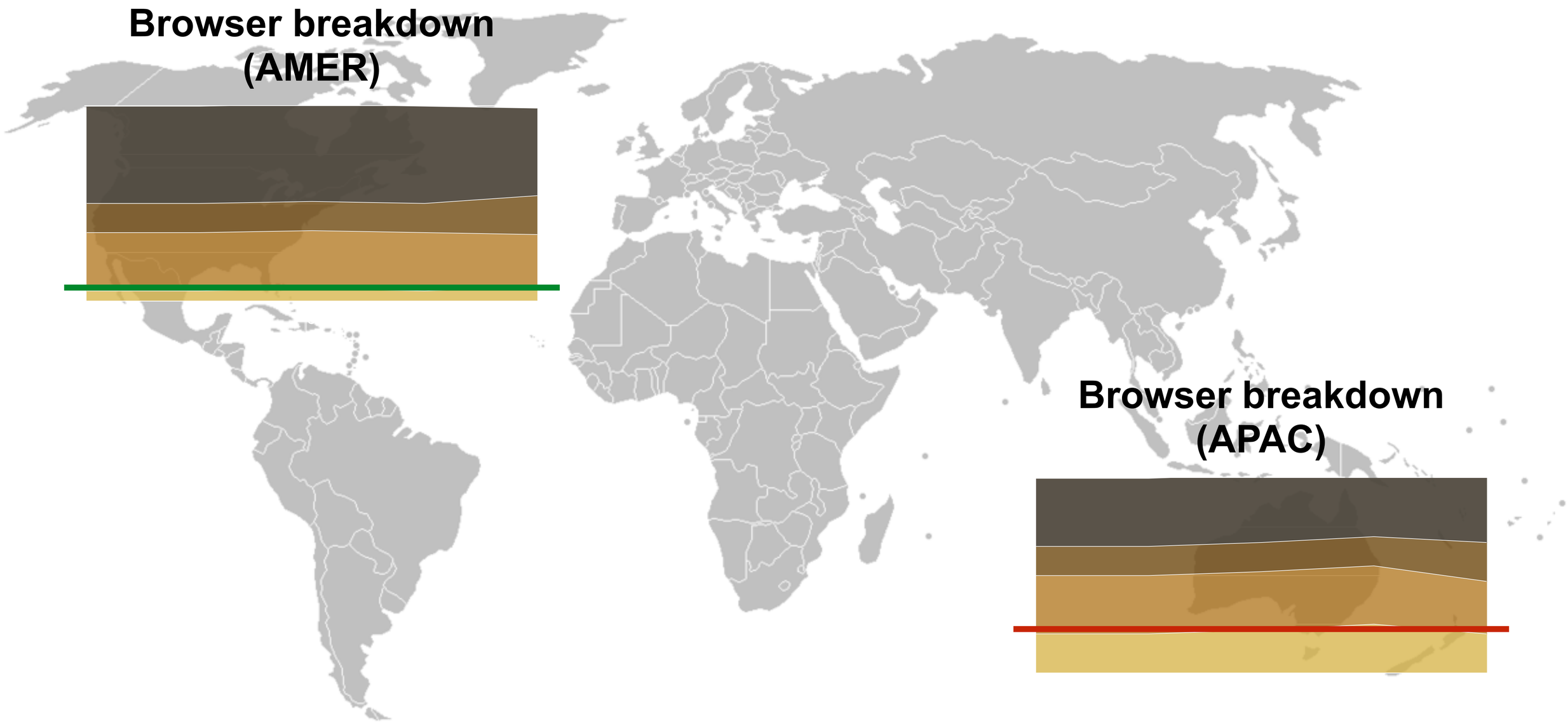
199 requests | 1.0 MB transferred | Finish: 35.36 s | Load: 4.76 s


Client
(Browser)
RUM agent
Service
APM agent

GET /order-coffee



Demo!


Browser breakdown
(AMER)
Browser breakdown
(APAC)

RUM: Going beyond the initial page load

WIP

- Route changes (Single-page apps)
- Interactions
- Performance marks

User defined performance marks

WIP

```
// Usage: performance.mark(name)

// The mark() method creates a timestamp in the
// browser’s performance entry buffer with the given name.
// The application defined timestamp can be retrieved by
// one of the Performance interface's getEntries*() methods.

performance.mark(heroBannerShown)
```

Transaction sample

Open in Discover

Sampled from 4800 - 5170 ms bucket

Timestamp	Duration	URL
1 hour ago (August 30th 2017, 14:16:15.883)	282 ms	https://www.opbeans.com/index

Timeline | System | User | Tags | Custom

/index

Hero ad banner complete
133 ms

Navigation timing · Resources

0 ms | 50 ms | 100 ms | 150 ms | 200 ms | 250 ms | 282 ms

DNS lookup

Connect

Sending and waiting for first byte

Downloading

Fetching, parsing and sync. execution

http://opbeans-node:3000/static/css/main.49b067e4.css

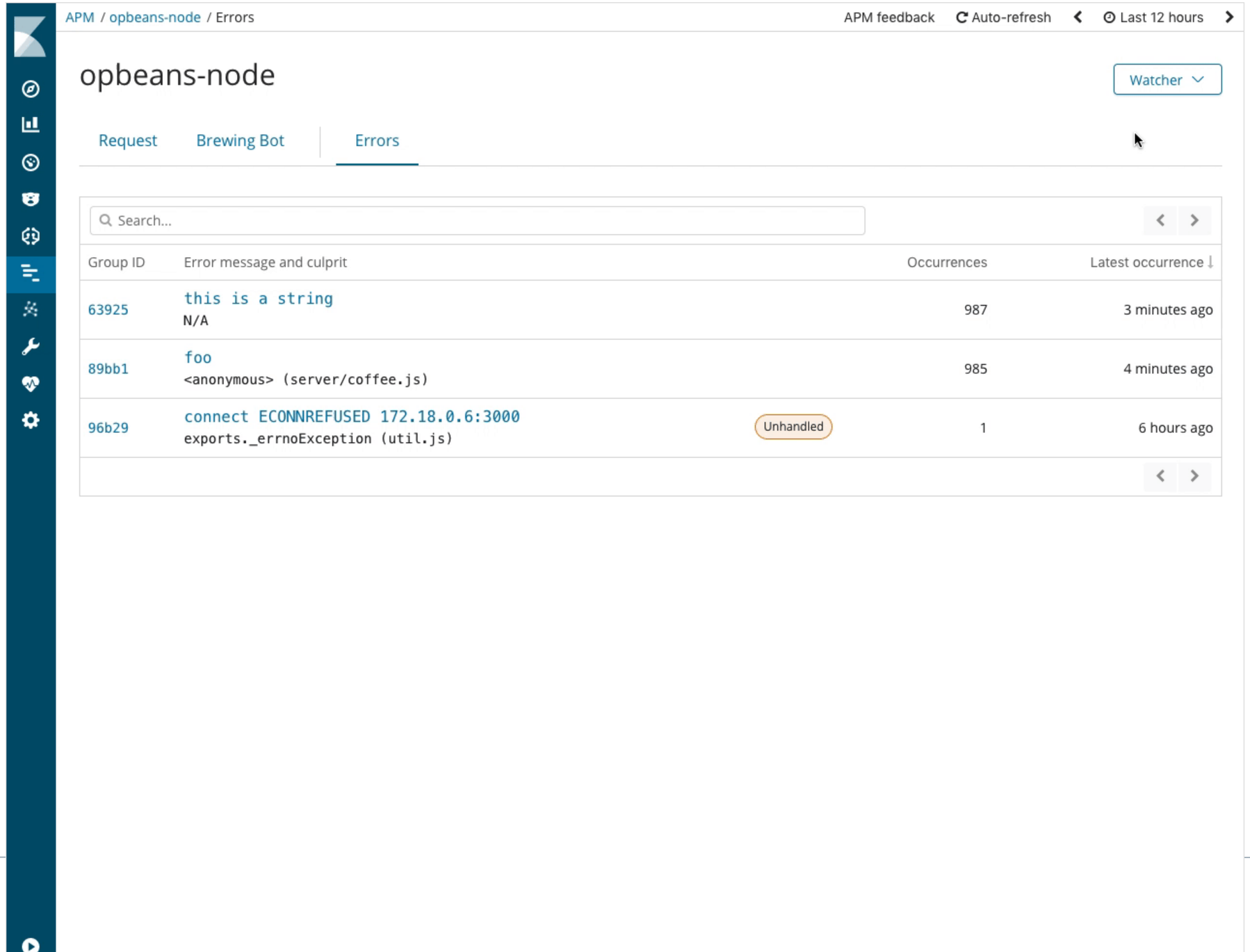
http://opbeans-node:3000/static/js/main.4531b054.js

http://opbeans-node:3000/static/media/icons.97493d3f.woff2

"DOMContentLoaded" event handling

"load" event handling

- Distributed tracing
- Machine Learning integration
- More agents
- Watcher integration


APM / opbeans-node / Errors
APM feedback
Auto-refresh
Last 12 hours
opbeans-node
Watcher
Request
Brewing Bot
Errors
Search...
Group ID
Error message and culprit
Occurrences
Latest occurrence
63925
this is a string
N/A
987
3 minutes ago
89bb1
foo
<anonymous> (server/coffee.js)
985
4 minutes ago
96b29
connect ECONNREFUSED 172.18.0.6:3000
exports._errnoException (util.js)
Unhandled
1
6 hours ago

Licensing

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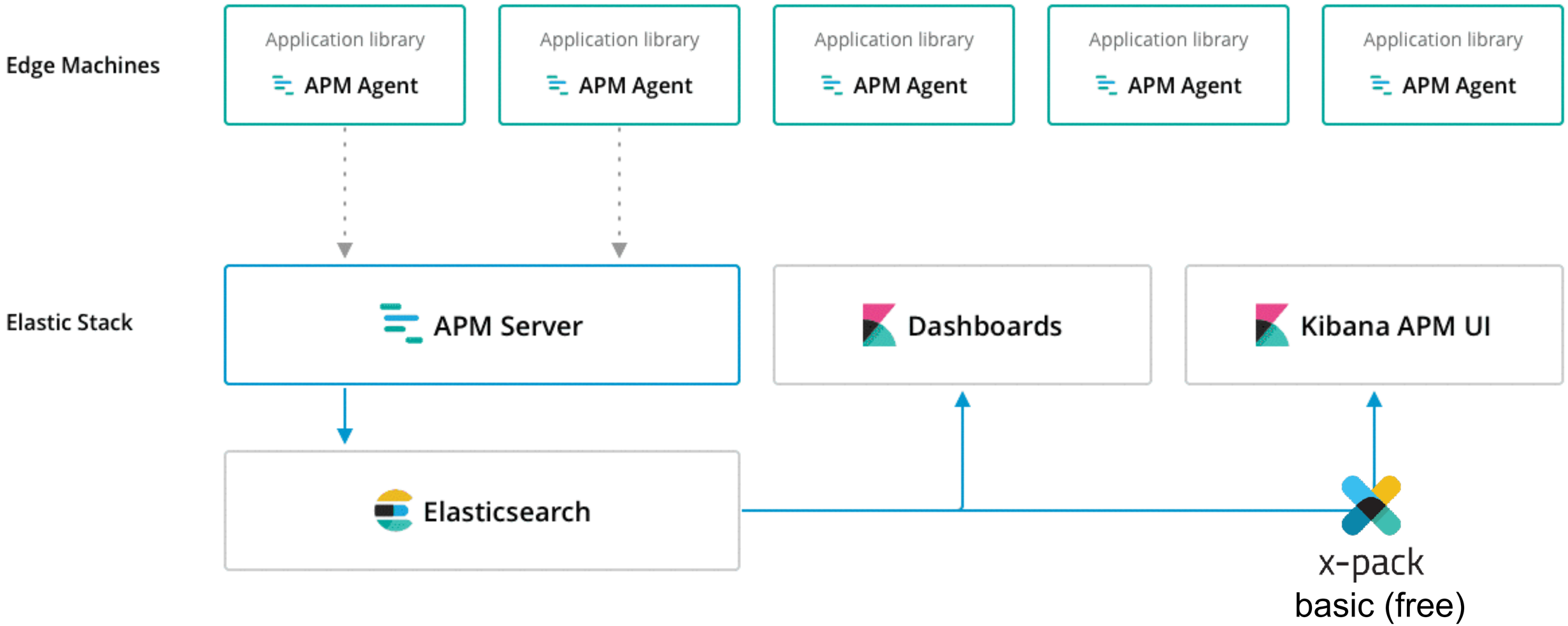
Solutions

Open Source Application Performance Monitoring

Already housing logs and system metrics in Elasticsearch? Expand to application metrics with Elastic APM. Four lines of code lets you see a bigger picture to quickly fix issues and feel good about the code you push.

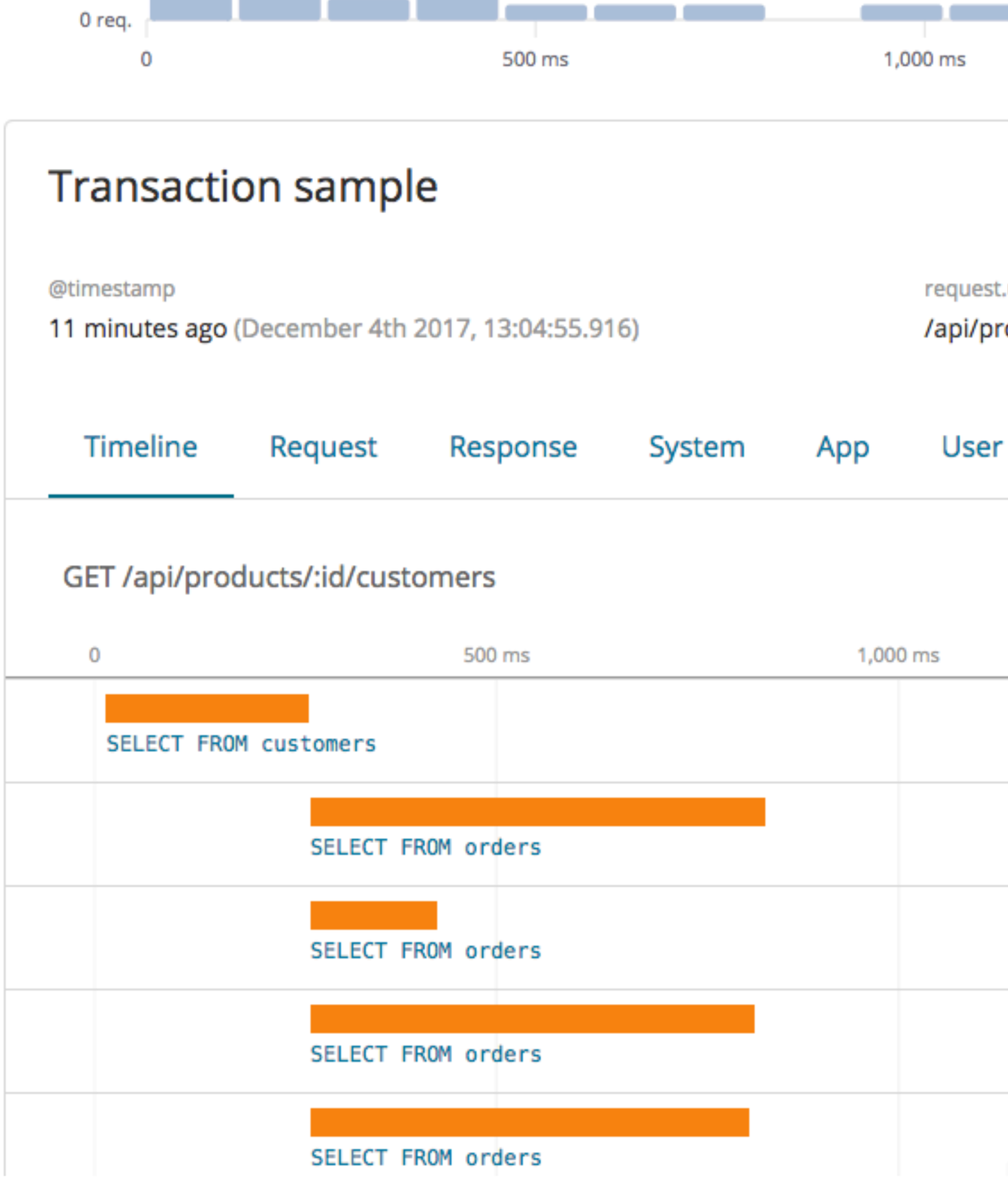
Open source APM



Supported agents

- Node.js
- Python
- Ruby

- RUM (Alpha)
- Java (WIP)
- Golang (WIP)



Elastic APM agents have built-in instrumentation

Subtitle

Our agents **automatically** instrument certain parts of the code, like database drivers, cache calls, template rendering, external HTTP requests, and more.

The agents are developed and maintained by our dedicated teams of language experts.

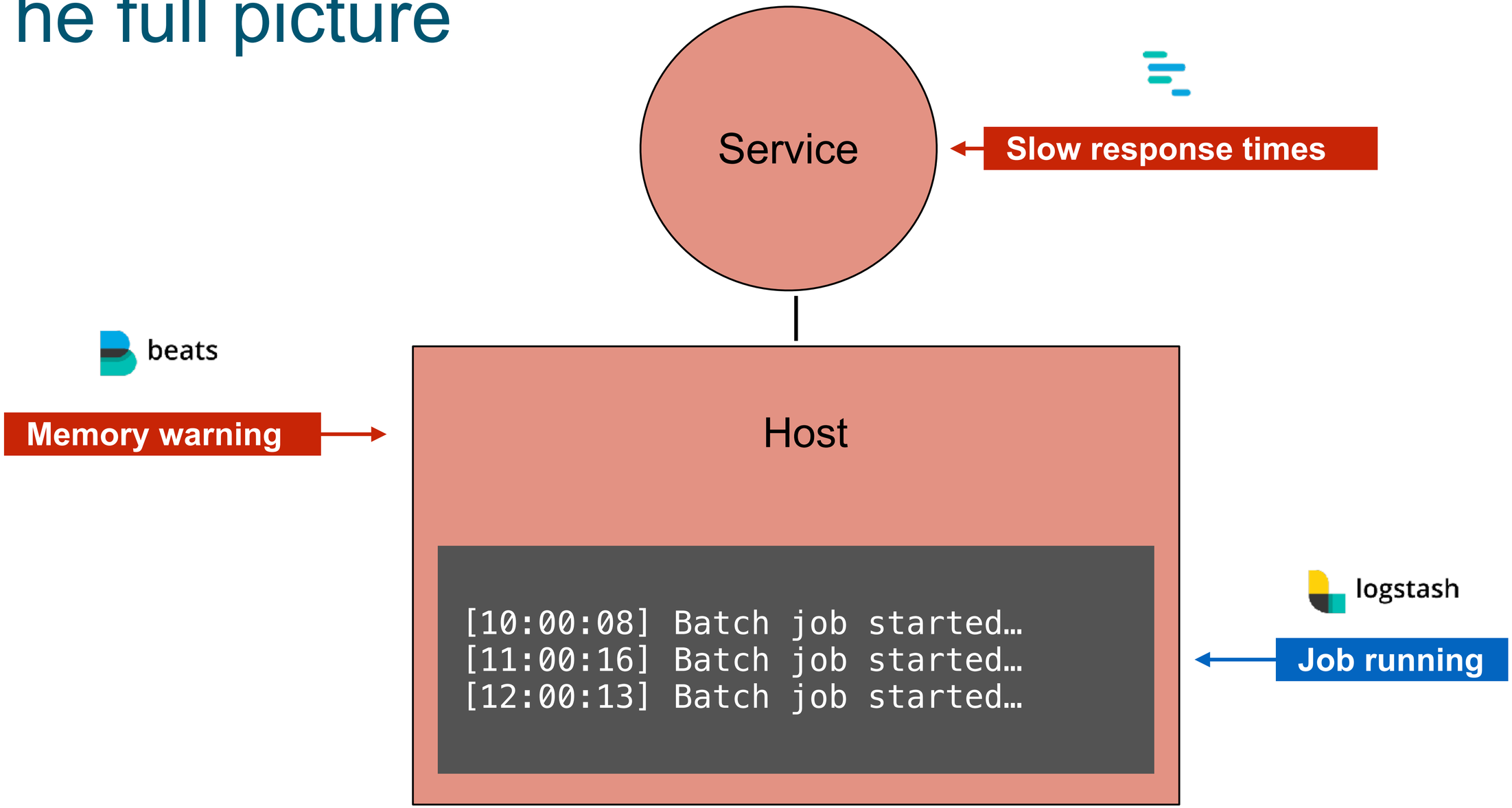
```
import elasticapm

@elasticapm.capture_span()
def coffee_maker(strength):
    fetch_water()

    with elasticapm.capture_span(‘near-to-machine’):
        insert_filter()
```

Just another index

The full picture



Using APM data for security

Subtitle

XSS

```
{ userId: 1510, sessionId: 12edS32res23ffu8 }
…
{ userId: 4609, sessionId: 12edS32res23ffu8 }
```

Watcher alert

SQLi

```
SELECT email, password, id FROM users WHERE email = '…'; UPDATE users SET email =
'heh@evilperson.com' WHERE email = 'admin@website.com'
```

Instrumenting Node.js Apps

Wednesday, 3:30pm

Thomas Watson
Node.js agent engineer,
APM, Elastic

Logs, Metrics and APM: The Holy Trinity of Operations

Thursday, 11:30am

Tanya Bragin
Senior Director of Product Management, Elastic

Get started today at elastic.co/downloads/apm

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Downloads

Download APM Server

Visit the APM board on Discuss and the agent repo on GitHub.

Version: 6.2.0

Release date: Februar 8

Notes: Elastic APM currently supports **Node.js**, **Python**, **Ruby**, and **JavaScript**.

Downloads:

DEB 32-BIT sha	DEB 6 ha	RPM 32-BIT sha
RPM 64-BIT sha	L ha	LINUX 64-BIT sha
MAC sha	WINDOWS 32-BIT sha	WINDOWS 64-BIT sha

elastic

www.elastic.co

Please attribute Elastic with a link to elastic.co

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